Make Better Use of Limited Tennis Training Hours

BUSINESS QUESTION

Which on-court performance measures should coaches prioritize to improve competitive outcomes?

22,883

ATP matches

2015–2023

Serve statistics • court surfaces • match outcomes

Decision-maker: Director of Player Development

GOAL | Turn completed-match statistics into practical coaching priorities.

Winning Is More About Serve Effectiveness Than Volume

Winner advantage in percentage points • 2015–2020 training seasons

First-serve points won

+10.24

percentage points

Second-serve points won

+10.84

percentage points

First serves in

+1.80

percentage points

The key distinction: getting a serve in is not the same as winning the point afterward.

COACHING LENS | Investigate serve placement, second-serve quality and the next shot.

A Transparent Model Matches Performance to Outcomes

2021 validation set • comparison against an uninformed baseline

Baseline

50.89%

validation accuracy

Logistic regression

93.82%

validation accuracy

Random forest

93.40%

validation accuracy

Why logistic regression?

Nearly the same accuracy with a clearer explanation of which performance measures matter.

HANDOFF TO MEEKI | Does the framework hold up across different court surfaces?

The Framework Shows Strong Results Across Three Court Surfaces

HARD COURT

93.1%

Accuracy

1,481 validation matches

F1: 0.9300

CLAY COURT

94.2%

Accuracy

641 validation matches

F1: 0.9415

GRASS COURT

96.6%

Accuracy

437 validation matches

F1: 0.9645

Consistent review principles across court surfaces.

A transparent framework can support coach judgment.

COACHING ACTION | Apply a consistent review process throughout the season.

Held-Out Season Testing Shows Promising Predictive Performance

93.13%

TEST ACCURACY

2022–2023 seasons
n = 5,411 matches

0.9822

ROC AUC

Strong ability to separate
match outcome groups

120 HOURS

ESTIMATED TIME TO REALLOCATE

Projected annual practice hours
per player

BASELINE ACCURACY **50.89%**

Model result is 42.24 percentage points higher.

The 120-hour figure is a projection; confirm real-world time savings during implementation.

Three Actions Can Turn Match Insights Into Better Training Decisions

01

SHIFT SERVICE DRILLS

Q1 implementation

Move 40% of weekly serve practice from first-serve volume toward placement depth and the next shot.

02

IMPROVE SECOND SERVES

Q1–Q2 focus

Practice kick variation, placement, and ways to counter aggressive returns—addressing the 10.8-point gap.

03

STANDARDIZE DEBRIEFS

Ongoing routine

Review match data within 12 hours and convert the main findings into goals for the next practice block.

FINAL RECOMMENDATION | Use match analytics to guide coaching—not replace coaching expertise.